



Goalie Rotation Timer

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2022 — Prototype

A small project that started as a timer to manage goalie rotations during our soccer games - everyone played as a keeper. The idea was simple: track playing intervals and use a buzzer to signal when it was time to switch keepers.

The first prototype was built around a single cylindrical battery at the center, with the modules arranged around it — an ESP32, a small display, and a charging board on different sides, and a buzzer mounted at the bottom. The second version moved to a custom PCB I designed, keeping the same core functionality but still in a prototype format.

Although the project was eventually scrapped, many of its ideas carried over into **NetScore**, which now handles score tracking through BLE button inputs. The keeper-switching logic might make a comeback there someday — though lately I've been playing more volleyball than football.

The prototype's code is available on [GitHub](#).

Notes:

- Two working prototypes: modular build + custom PCB
- ESP32-based with display, buzzer, and battery management
- Designed to automate goalie rotation timing
- Later merged conceptually into **NetScore**

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2022 / *Embedded Systems, Hardware Prototyping*

A portable, ESP32-based timer with a display and buzzer, prototyped to automate goalie rotation intervals during soccer games.

(Here you should include pictures of both prototypes: the modular one built around the battery and the custom PCB version.)

The Story

This project started as a dedicated device to manage goalie rotations during our soccer games, where everyone had to play as a keeper. The idea was simple: track a set time interval for each player and sound a buzzer when it was time to switch. I built two functional prototypes. The first was a modular build centered around a cylindrical battery, with an ESP32, a small display, and a charging board on different sides. The second version advanced to a custom PCB I designed, keeping the same core functionality in a more integrated format.

Key Features

- **Automated Rotation Timing:** Tracks playing intervals and provides audible alerts for substitutions.
- **ESP32 Core:** Utilizes an ESP32 for all timing, display, and alert logic.
- **Integrated Hardware:** Features a small display, a buzzer, and a self-contained battery management system.
- **Iterative Prototyping:** Evolved from a modular, hand-wired build to a custom-designed PCB.

Tech Stack

- **Microcontroller:** ESP32
- **Hardware:** Custom PCB, display module, buzzer, Li-ion battery charging circuit.
- **Programming:** C++ (Arduino Framework)

Challenges & Lessons Learned

While the project was eventually scrapped as a standalone device, it was an invaluable exercise in rapid hardware prototyping and product design. The process of building two distinct versions taught me a lot about component selection and PCB layout for compact devices. More importantly, its core concepts directly inspired and carried over into my later NetScore project, demonstrating how ideas from one prototype can seed a more ambitious successor.

Links

- GitHub Repository.